

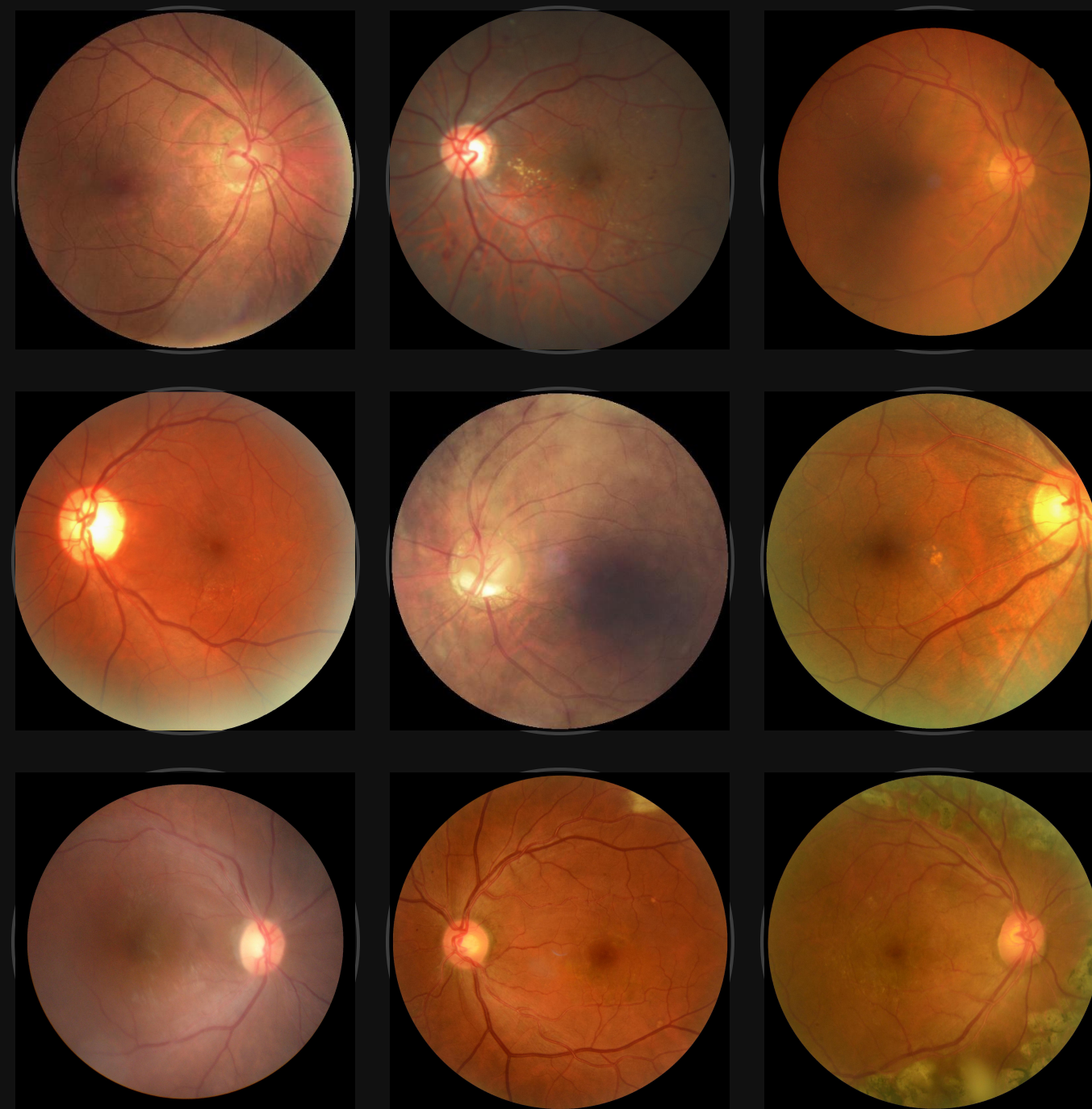
TEAM
EyeQ

SPRINT 0

DUAL

OCULAR DISEASE SCREENER

AI-assisted screening for cataract and glaucoma from a single retinal photograph per eye.





What we'll cover

01 Team roles and responsibilities

02 Problem statement and hypothesis

03 Project description

04 Personas

05 Technologies

06 Algorithms

07 Project schedule

08 Team working agreement

09 Retrospective and wiki



Roles & responsibilities



Anel Bazarbayeva

Scrum Master, MLOps/
Integration Developer



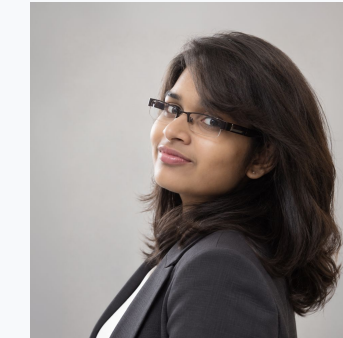
Negar Kamvij

ML Developer, External
Validation & Calibration



Shridhi Patel

Team Leader, ML Developer,
Glaucoma Lead & Data
Pipeline Developer



Bhoami Khona

Frontend Developer



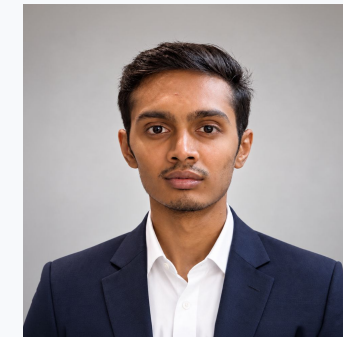
Krushiben Patel

Data Pipeline Developer



Ansh Kevadiya

ML Developer,
Cataract Lead



Smit Meghapara

ML Developer,
Interpretability & Safety



Problem statement

PROBLEM STATEMENT

Cataract and glaucoma cause a large share of the world's vision loss, but they fail patients in opposite ways: cataract is treatable if caught, glaucoma has no symptoms until the damage is already permanent. Both require routine screening that many at-risk patients never get.

EVIDENCE TO CITE

45%

of global blindness caused by cataract (WHO / GBD 2020)

Half

of US glaucoma cases go undiagnosed (CDC)

0.90 → 0.58

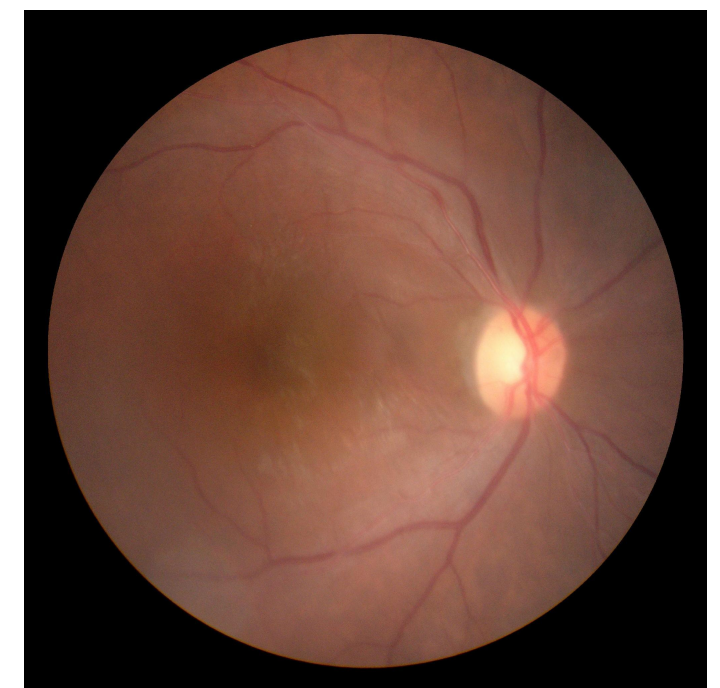
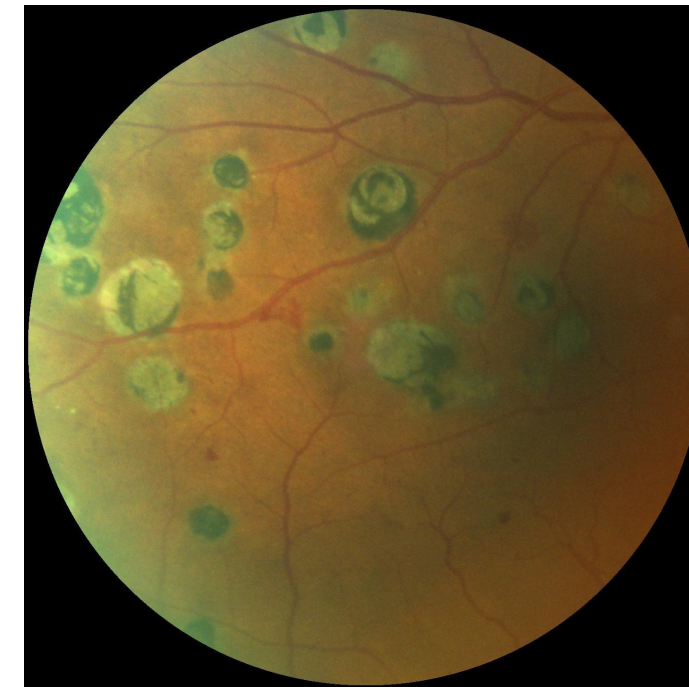
AI screeners: AUC in-house drops across sites (J Clin Med, 2026)



Hypothesis

We believe that giving primary care and community screening sites a single-photo AI tool that flags both cataract and glaucoma risk will result in more at-risk patients getting referred before damage is done.

We will know we are right when the model holds up not just on our own test data, but on photos from different fundus cameras and patient populations it never trained on.

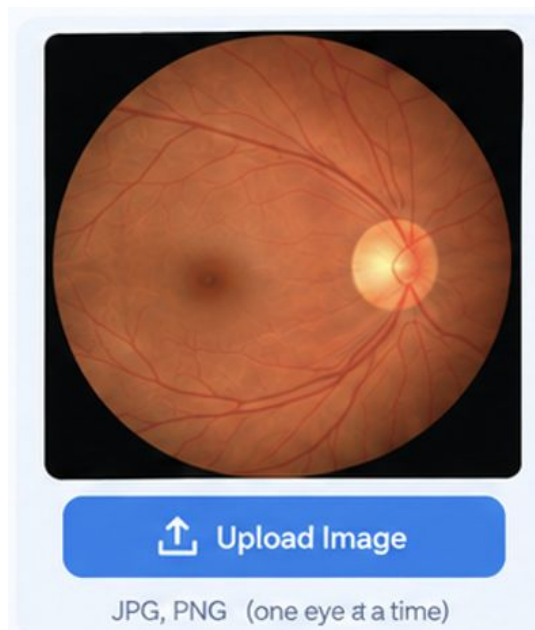




What we are building

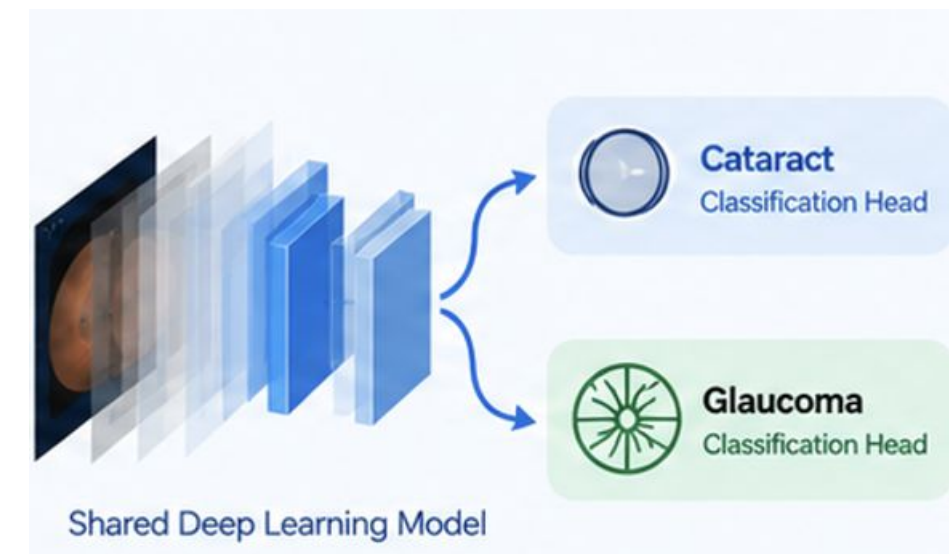
Dual is a web-based AI screening application that analyzes a fundus photograph of the eye and provides separate risk results for **cataract and glaucoma**, with confidence information and a referral recommendation when the image or prediction is uncertain.

01 Upload Fundus Image



One retinal photograph per eye is uploaded to Dual.

02 AI Analysis

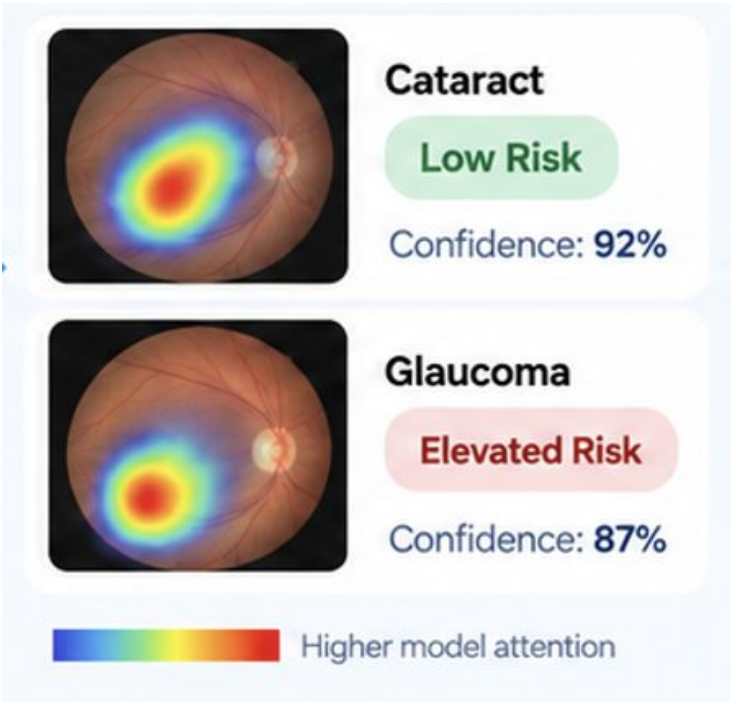


One shared deep learning model analyzes the image for both cataract and glaucoma.



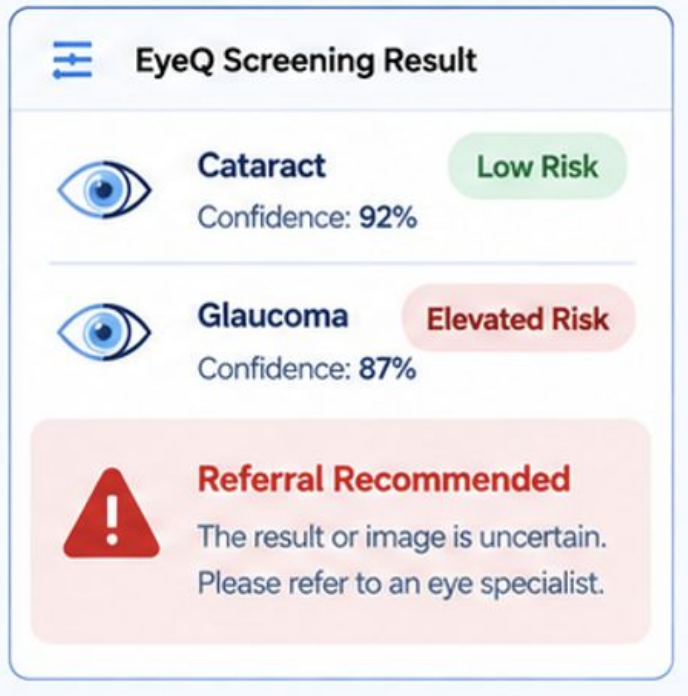
Dual Screening Result

03 Risk Results



Dual provides separate risk results with confidence levels and highlighted regions in the image.

04 Referral Recommendation



Dual provides separate risk results with confidence levels and highlighted regions in the image.

Scope: Dual supports screening and referral decisions. It does not provide medical diagnosis, treatment advice, EHR integration, or other eye conditions.

**CLEARER VISION
HEALTHIER LIVES**



“I want to keep seeing the moments that matter.”

01 | PATIENT PERSONA



Helen Whitfield, 68

Retired · At-risk older adult

Last eye exam four years ago

Wants to stay proactive about eye health

GOALS

- Identify possible cataract or glaucoma concerns early
- Understand her eye health clearly
- Know what to do next

NEEDS

- Plain-language results
- Large, easy-to-read text
- A clear next step

CONCERNS

- Confusing medical terminology
- A frightening result without explanation
- Not knowing who to ask for help

“I may not have symptoms now, but I want to understand my screening result and what I should do next.”

SCENARIO

Helen attends a screening visit. A technician captures her fundus image and runs DUAL. She receives a simple explanation of the screening result and the recommended next step. DUAL supports screening and referral; it does not provide a diagnosis.



“I want to help more people get screened, quickly and accurately.”

02 | SCREENING TECHNICIAN PERSONA



Arun Desai, 26

Screening Technician

Works at rural and community health camps

Uses a portable retinal/fundus camera

Focused on efficient, reliable screening

GOALS

- Capture high-quality images quickly
- Get easy-to-understand screening results
- Screen patients efficiently

NEEDS

- Immediate image-quality feedback
- Simple, step-by-step workflow
- Clear next-step guidance

CONCERNS

- Poor-quality or ungradable images
- Patients leaving before results are ready
- Misinterpreting screening results

“I need clear feedback right away so I can move to the next patient without delays.”

SCENARIO

Arun captures a patient’s fundus image at a screening camp and uploads it to the DUAL. The system checks image quality and screens for cataract and glaucoma. Arun receives a clear screening result and recommended next step and can generate a report for clinical review when needed.



“I want a quick, reliable way to know which patients may need specialist review.”

03 | PRIMARY CARE PHYSICIAN PERSONA

06 • PERSONAS



Dr. Maya Rao, 41

Primary Care Physician

Family physician in a busy primary care practice

No ophthalmology specialization

Uses screening information to support referral decisions

GOALS

Identify patients who may need specialist evaluation

Support early detection and preventive care

Review screening information quickly

NEEDS

Clear cataract and glaucoma screening results

Clear risk and uncertainty indicators.

A clear recommended next step

CONCERNS

Missing a patient who may need referral

Over-referring patients unnecessarily

Misinterpreting an uncertain AI result

“I need a clear screening result that helps me decide when further eye-care evaluation may be appropriate.”

SCENARIO Dr. Rao reviews a DUAL screening report after a technician captures and analyzes a patient’s fundus image. She checks the cataract and glaucoma screening results, uncertainty, and recommended next step, then decides whether further eye-care evaluation is appropriate. DUAL supports screening and referral; it does not diagnose.

DUAL in Action

Early Detection. A Brighter Tomorrow. 

01 CAPTURE



“Time for my eye screening.”
Helen — Patient



“Let me capture your eye image.”
Arun — Screening Technician

02 SCREEN



DUAL



Checking image...

Screening Cataract +
Glaucoma...

Screening complete!

✓ **Cataract**

No significant signs detected

⚠ **Glaucoma**

Possible elevated risk (Refer for further evaluation)

03 REVIEW



“I’ll review the result.”

Dr. Maya — Primary Care Physician

04 NEXT STEP



“Now I know my next step.”

Helen — Patient

Tech Stack

01

Model / ML

Computer vision and deep learning for robust, clinically relevant predictions.

Python

PyTorch

OpenCV

Weights & Biases

02

Backend

High-performance APIs and data services for secure, reliable inference.

FastAPI

PostgreSQL

Object Storage

03

Frontend

A responsive, clinician-friendly interface for seamless screening workflows.

HTML

CSS

JavaScript

React

Next.js

04

Infra & Tooling

Scalable infrastructure and modern developer tooling for iteration and deployment.

Docker

GitHub

GitHub Actions

Vercel

Render

Cloud GPU



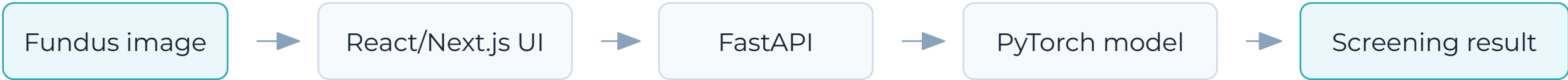
TRAINING & VALIDATION DATA

Train: **ODIR-5K**

External validation / test-only: ORIGA · DRISHTI-GS · ACRIMA · retina_dataset_2016

REAL DATA
STRONGER MODELS
BETTER OUTCOMES

END-TO-END PIPELINE



FROM IMAGE
TO INSIGHT
FOR BRIGHTER LIVES



Approach & evaluation

CANDIDATE APPROACHES

- 01 CNN transfer learning**
Use ResNet/EfficientNet to learn disease patterns from eye images.
- 02 Multi-task heads**
One model detects both cataract and glaucoma using shared features.
- 03 Baseline for comparison**
Use Logistic Regression as a simple model for comparison.
- 04 Image-quality classifier**
Detects ungradable or poor-quality eye images before prediction

HOW WE WILL JUDGE IT

Sensitivity

How well the model detects actual cataract/glaucoma cases.

Specificity

How well it identifies healthy eyes.

AUC (Area Under the Curve)

Overall ability to distinguish between healthy and diseased eyes.

Confusion Matrix

Shows correct and incorrect predictions for each condition.

Test Set

Evaluate on unseen patient images to measure real-world performance.

Key goal • Minimize missed disease cases while maintaining reliable predictions.



Cadence & deliverables

MEETING CADENCE

Standup — Quick progress discussion after class.

Working session — Sunday 2-hour Zoom meeting + individual work throughout the week.

Sprint Duration — Approximately 4 weeks.

SPRINT DELIVERABLES

- **Sprint 0 — SEP-1 to SEP-21**
Finalize the project scope, team responsibilities, personas, technology stack, datasets, and overall development plan.
- **Sprint 1 — SEP-21 to OCT-19**
Prepare and preprocess retinal images, create the basic user interface, and build initial cataract and glaucoma model functionality.
- **Sprint 2 — OCT-19 to NOV-16**
Develop the shared dual-head model, evaluate both disease predictions, integrate the model with the application, and begin external validation.
- **Sprint 3 — NOV-16 to DEC-14**
Complete the end-to-end application, perform final model testing and validation, improve the interface, and finalize project documentation and presentation



How we work together

Where & when we meet

Two-hour Zoom meeting every Sunday for project work and planning. Brief follow-up discussions after Tuesday's class.

Communication

For Primary channel, use the team group chat for regular updates and coordination. Respond within 24 hour.

If someone doesn't commit

Talk with the member first and work together to resolve the issue. Repeated problems will be discussed as a team and, if necessary, brought to the instructor.

Deadlines & reminders

Set internal deadlines ahead of official due dates. The team leader sends reminders before deadlines. Work is considered complete when it is finished, tested, reviewed, and submitted to the team repository.



Sprint 0 retro

What went well

We successfully defined the Dual Ocular Disease Screener and its scope.

We identified our target users/personas and initial technology/algorithm approach.

Team members divided Sprint 0 responsibilities across the presentation deliverables.

What can improve

We could communicate more consistently when tasks or plans change.

We could assign responsibilities earlier so everyone knows their focus from the start.

We could review the dataset and technical approach sooner to make the transition into Sprint 1 smoother.

ACTION 1 → SPRINT 1 BACKLOG

Finalize Sprint 1 task ownership and deadlines.

Owner: Team Lead + all team members

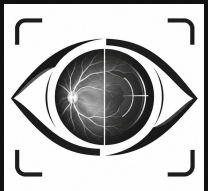
By when: Before Sprint 1 development begins.

ACTION 2 → SPRINT 1 BACKLOG

Review the dataset and establish the initial data pipeline/model baseline.

Owner: Assigned ML/Data team members

By when: During Sprint 1.

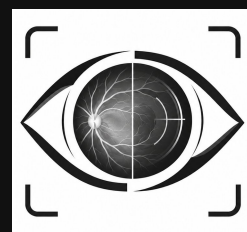


GROUP WIKI

<https://github.com/htmwf2026-Team1/wiki>

Earlier Detection. Brighter Tomorrows.





Thank you

TEAM EYEQ